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Smartphone battery behavior is notoriously unpredictable. Devices that last an entire day under light use may drain rapidly under heavy activity, and phones often shut down unexpectedly despite displaying a nonzero battery percentage. These behaviors arise from the interaction between user activity, power-hungry hardware subsystems, battery aging, environmental temperature, and voltage limitations intrinsic to lithium-ion batteries.

To address this problem, we develop a continuous-time, physics-based smartphone battery model that links usage patterns directly to battery depletion through conservation of charge and electrical principles. The model tracks the battery state of charge (SOC) using a differential equation driven by current draw, while power demand is modeled as the sum of interpretable subsystem contributions, including screen brightness, CPU load, network activity, GPS usage, and background processes. Crucially, the framework incorporates terminal voltage behavior, recognizing that device shutdown occurs when voltage under load falls below a critical cutoff rather than when stored charge is fully depleted.

Battery aging is represented through an effective capacity and internal resistance derived from reported battery health (SOH), while environmental temperature modifies both parameters to capture well-known cold-weather effects. To account for unobserved variability in usage and power demand, the model integrates a Hidden Markov Model (HMM) for discrete usage modes and an Extended Kalman Filter (EKF) that fuses voltage measurements with SOC dynamics for improved state estimation.

The model is validated using real-world smartphone sensing data and reproduces key qualitative features of observed battery behavior. Predicted time-to-empty increases monotonically with SOC, while dispersion across usage scenarios reflects realistic variability in power demand. The model clearly distinguishes between SOC-limited and voltage-limited shutdown regimes and captures increased sensitivity to temperature and load at low SOC.

Scenario-based simulations quantify battery life across common usage patterns (idle, reading, video streaming, navigation, and gaming) and environmental conditions. A sensitivity analysis identifies temperature, sustained power demand, and voltage sag as the dominant drivers of battery life. Low-power scenarios exhibit the largest runtime loss per degree Celsius, while high-power scenarios become voltage-limited earlier and shut down at higher SOC.

These findings lead to actionable recommendations. At the user level, reducing CPU-intensive tasks, limiting unnecessary network activity, and disabling GPS when not required yield the largest gains in battery life. At the system level, operating systems can improve reliability through voltage-aware SOC estimation, temperature-adaptive power management, and proactive mitigation of voltage-limited shutdown.

Overall, this framework provides a physically interpretable, extensible, and realistic model of smartphone battery behavior that explains common user-observed phenomena and supports informed decision-making to extend battery life.

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1 Introduction

Smartphones have become indispensable tools in modern life, supporting communication, navigation, entertainment, and productivity for billions of users worldwide. Despite substantial advances in hardware efficiency and battery technology, unpredictable battery behavior remains a persistent and widely experienced problem. A device that lasts an entire day under light use may discharge rapidly under heavy activity, and phones frequently shut down unexpectedly despite displaying a nonzero battery percentage. Such behavior can be especially disruptive during critical tasks such as navigation, emergency communication, or travel in unfamiliar environments.

These inconsistencies arise from the complex interaction between user behavior, hardware subsystems, environmental conditions, and the internal electrical state of the battery. While battery percentage is commonly interpreted as a direct indicator of remaining energy, device operability is in fact governed by terminal voltage, not stored charge alone. High power demand, increased internal resistance due to aging, and low temperatures can all cause the battery voltage to fall below a critical cutoff even when substantial charge remains, leading to early shutdown.

In this report, we develop a continuous-time mathematical model of smartphone battery behavior grounded in physical principles governing lithium-ion cells. The model tracks the battery state of charge through conservation of charge while explicitly accounting for usage-dependent power demand from major phone subsystems such as the display, processor, network interfaces, and GPS. A voltage-based constraint is incorporated to capture load-dependent voltage sag and to explain early shutdown behavior that cannot be explained by state of charge alone.

Battery aging is represented through effective capacity and internal resistance parameters consistent with battery health metrics reported by modern smartphones. Environmental effects, particularly temperature-dependent changes in capacity and resistance, are also included, enabling the model to capture the pronounced degradation in performance observed in cold conditions. To improve realism under variable usage, voltage measurements are used to guide state estimation, and usage patterns are represented through interpretable power scenarios.

Using this framework, we simulate battery discharge under a range of realistic usage, aging, and environmental conditions. We evaluate time-to-empty predictions, identify the dominant contributors to rapid battery drain, and quantify sensitivity to temperature and power demand. These results are then translated into practical recommendations for smartphone users and insights into how operating systems can implement more effective, voltage-aware power management strategies.

2 Assumptions and Model Overview

2.1 Modeling Approach and Scope

We model smartphone battery behavior using a continuous-time, physics-based framework grounded in conservation of charge and basic electrical principles. The battery is treated as a dynamical system

whose physical constraints determine how the state of charge evolves in response to time-varying power demand from the device.

Rather than relying on purely empirical curve fitting or black-box prediction methods, the structure of the model is derived from physical reasoning. Empirical data, manufacturer specifications, and reported battery health metrics are used only to estimate parameters and assess plausibility. This separation between model structure and parameter estimation preserves interpretability and allows the framework to explain observed phenomena such as rapid discharge or early shutdown, rather than merely reproducing them.

The model operates at the device level rather than at the electrochemical scale. Detailed internal battery chemistry is abstracted into a small number of effective parameters, enabling realistic predictions while maintaining analytical clarity and computational tractability.

2.2 Core Assumptions

The following assumptions are adopted throughout the model:

- The smartphone is powered by a single lithium-ion battery and is modeled in continuous time, as specified in the problem statement.
- The state of charge (SOC) represents the fraction of total available charge remaining in the battery.
- The battery open-circuit voltage is almost a monotone function of SOC, ranging approximately from 4.2 V at full charge to 3.0 V near depletion; device shutdown occurs when the terminal voltage falls below a fixed cutoff, even if SOC is nonzero.
- Battery aging is captured through a reduced effective capacity, consistent with battery health indicators reported by modern smartphones.
- Internal resistance is treated as constant within a given scenario but may vary across scenarios representing different battery ages or environmental conditions.
- Temperature effects are incorporated as scenario-based modifiers to capacity and internal resistance rather than through a fully coupled thermal model.
- Power consumption from individual subsystems can be aggregated into a total time-dependent power demand determined by usage patterns.

These assumptions allow the model to capture the dominant drivers of battery depletion and shutdown behavior while avoiding unnecessary complexity.

2.3 Model Architecture

The model links user behavior to battery depletion through a sequence of physically meaningful relationships. User and operating-system controls determine the instantaneous power demand of the smartphone, which is modeled as the sum of contributions from major subsystems such as the display, processor, network interfaces, GPS, and background processes. This aggregate power demand $P(t)$ determines the electrical current drawn from the battery.

The battery state of charge (SOC) evolves continuously according to this current draw and the battery's effective capacity. At the same time, the terminal battery voltage is computed as a function of SOC and internal resistance, capturing load-dependent voltage sag. Device shutdown is modeled as a voltage-based event, occurring when the terminal voltage under load falls below a fixed cutoff threshold. This mechanism explains commonly observed behavior such as phones shutting down at low but nonzero displayed battery percentages, particularly under high-load conditions or with aged batteries.

By explicitly modeling the relationships between power demand, current, charge depletion, and voltage constraints, the framework supports realistic time-to-empty prediction, sensitivity analysis, and scenario-based comparisons across usage patterns, battery health, and environmental conditions.

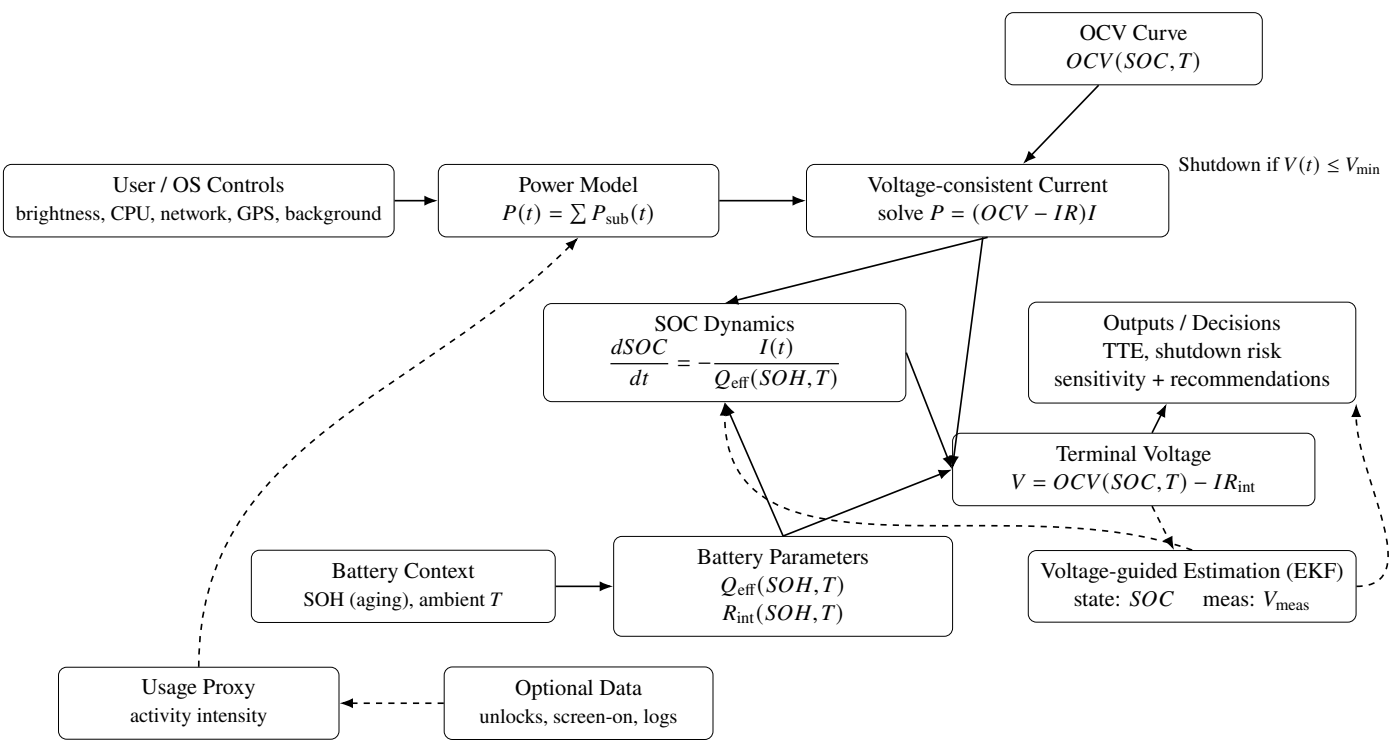


Figure 1: Clean model architecture. Solid arrows denote physical causality; dashed arrows denote inferred or estimation-driven influence. User behavior determines power demand, which drives current, SOC depletion, and voltage sag. Voltage governs shutdown and guides SOC estimation. Outputs include time-to-empty, sensitivity, and recommendations.

3 Continuous-Time Battery Model

Each parameter in the model is defined in the Appendix 13.1.

3.1 State of Charge Definition and Dynamics

We define the battery state of charge (SOC) as the fraction of effective charge remaining:

$$SOC(t) \;=\; \frac{Q(t)}{Q_{\text{eff}}}, \qquad 0 \le SOC(t) \le 1, \tag{1}$$

where $Q(t)$ is the remaining charge (Ah) and Q_{eff} is the effective battery capacity (Ah). Battery aging is incorporated by allowing Q_{eff} to vary across scenarios (or slowly over long time scales).

Using conservation of charge (coulomb counting), SOC evolves according to

$$\frac{dSOC}{dt} \;=\; -\frac{\eta(t)\, I(t)}{Q_{\text{eff}}}, \tag{2}$$

where $I(t)$ is the battery current (A), defined positive when discharging, and $\eta(t)$ is the Coulomb efficiency[1]. For lithium-ion batteries during discharge, $\eta(t) \approx 1$, so we take $\eta(t) = 1$ unless stated otherwise.

Equation (2) is the primary continuous-time ODE governing battery depletion. All additional model components (power demand, voltage response, shutdown behavior, and state estimation) exist to determine (or constrain) the current $I(t)$ and to improve inference of $SOC(t)$ from observable quantities.

3.2 Power Demand and Battery Current

Smartphone usage is more naturally described in terms of power demand $P(t)$ (W). Battery current and power are related by

$$P(t) \;=\; V(t)\, I(t), \tag{3}$$

so that

$$I(t) \;=\; \frac{P(t)}{V(t)}. \tag{4}$$

Because the terminal voltage $V(t)$ depends on current through internal resistance, (4) is implicit and must be resolved jointly with a voltage model (Section 4).

[1] Prarthana Pillai, James Nguyen, and Balakumar Balasingam. *Performance Analysis of Empirical Open-Circuit Voltage Modeling in Lithium Ion Batteries, Part-1: Performance Measures*. 2023. arXiv: 2306.16542 [eess.SY]. URL: https://arxiv.org/abs/2306.16542.

3.3 Power Decomposition by Subsystem

We model total smartphone power demand as a sum of interpretable subsystem contributions:

$$P(t) = P_{\text{idle}} + P_{\text{screen}}(t) + P_{\text{cpu}}(t) + P_{\text{net}}(t) + P_{\text{gps}}(t) + P_{\text{bg}}(t). \tag{5}$$

Here P_{idle} represents baseline system power, while the remaining terms capture incremental power consumption due to specific user activities. This decomposition supports scenario simulation and attribution of battery drain to individual behavioral drivers.

Screen power. Let $s(t) \in \{0, 1\}$ indicate whether the screen is on and let $b(t) \in [0, 1]$ denote normalized brightness. We model screen power as

$$P_{\text{screen}}(t) \;=\; s(t)\big(a_0 + a_1\, b(t)^{\beta}\big), \tag{6}$$

where a_0 captures fixed screen-on overhead, a_1 scales with brightness, and $\beta > 0$ allows for nonlinear brightness–power behavior.

CPU / application power. Let $\ell(t) \in [0, 1]$ denote normalized processor load, reflecting the intensity of active applications (e.g., messaging, browsing, gaming, or video playback). We model compute power as

$$P_{\text{cpu}}(t) \;=\; c_0 + c_1\, \ell(t)^{\gamma_c}, \tag{7}$$

where $\gamma_c \geq 1$ permits superlinear power growth under heavy computation. Different usage scenarios correspond to different typical values and time-profiles of $\ell(t)$.

Network power. Network draw depends on radio type and signal conditions. Let $m(t)$ denote the active network mode (e.g., WiFi, LTE/5G, weak-signal cellular), encoded by indicator functions $\mathbf{1}_{m(t)=k}$:

$$P_{\text{net}}(t) \;=\; \sum_{k \in \mathcal{M}} n_k\, \mathbf{1}_{m(t)=k} \;+\; n_u\, u(t), \tag{8}$$

where n_k represents baseline power for mode k and $u(t)$ is an optional traffic-intensity proxy (e.g., sustained throughput).

GPS and background activity. GPS power is modeled as

$$P_{\text{gps}}(t) \;=\; g(t)\, g_0, \tag{9}$$

where $g(t) \in \{0, 1\}$ indicates GPS activity. Background power is modeled as

$$P_{\text{bg}}(t) \;=\; b_0 + b_1\, r(t), \tag{10}$$

where $r(t) \in [0, 1]$ captures background synchronization intensity and system task activity.

These low-dimensional subsystem models balance realism and interpretability while enabling parameter estimation, sensitivity analysis, and actionable recommendations.

3.4 Role of the Continuous-Time Model

Together, equations (2)–(5) define a continuous-time, usage-driven battery depletion model in which user behavior enters exclusively through power demand, while physical battery constraints govern how that demand translates into current draw and charge loss. This separation between behavioral inputs and physical dynamics is a central design feature of the framework.

The power decomposition in (5) enables realistic scenario construction (e.g., gaming, navigation, video streaming, or idle use) while preserving interpretability: each subsystem contribution can be independently varied to assess its impact on battery lifetime. At the same time, the SOC-ODE (2) ensures that all scenarios obey charge conservation.

Crucially, the current $I(t)$ appearing in (2) cannot be determined from power demand alone. Because terminal voltage depends on SOC, temperature, internal resistance, and load, the relationship between $P(t)$ and $I(t)$ must be resolved consistently with a voltage model. This coupling is essential for capturing voltage-limited shutdown behavior observed in real devices, where a phone may power off before all stored charge is depleted.

In the following section, we introduce a voltage response model that links SOC and current to terminal voltage, define a voltage-based shutdown criterion, and show how voltage measurements can be used to guide state-of-charge estimation. These components complete the physical description of battery behavior and enable realistic time-to-empty predictions under diverse operating conditions.

4 Voltage Response, Shutdown, and State Estimation

4.1 Terminal Voltage Model and Shutdown Criterion

Although the internal circuitry of smartphones is complex, battery behavior at the device level can be accurately captured using an equivalent circuit model (ECM). We adopt a first-order Thevenin ECM, which provides a favorable balance between physical interpretability, numerical stability, and modeling simplicity. Under load, the terminal voltage of a lithium-ion battery is modeled as

$$V(t) = V_{\mathrm{OCV}}(SOC(t), T) - I(t)\, R_{\mathrm{int}}(T), \tag{11}$$

where $V_{\mathrm{OCV}}(SOC, T)$ denotes the open-circuit voltage as a function of state of charge and ambient temperature T, and $R_{\mathrm{int}}(T)$ is an effective internal resistance.

This formulation captures voltage sag arising from internal losses, which becomes pronounced under high current draw, low temperature, or battery aging. Rather than employing detailed electrochemical voltage models (e.g., Shepherd-type models), we represent $V_{\mathrm{OCV}}(SOC, T)$ using a smooth, monotone approximation. Such representations are standard in battery management systems and provide sufficient accuracy for voltage-based estimation without unnecessary complexity.

Device shutdown is modeled as a voltage threshold event:

$$\text{shutdown occurs if} \quad V(t) \le V_{\min}, \tag{12}$$

where $V_{\min}$ is the minimum operating voltage of the device. Because shutdown depends on terminal voltage rather than SOC directly, a phone may power off at a nonzero displayed battery percentage, particularly under high load, increased internal resistance, or cold environmental conditions.

4.2 Consistent Current Computation

Battery current and power demand are related through $P(t) = V(t)I(t)$. Substituting (11) yields

$$P(t) = V_{\mathrm{OCV}}(SOC(t), T)\, I(t) - R_{\mathrm{int}}(T)\, I(t)^2. \tag{13}$$

For known $SOC(t)$ and power demand $P(t)$, this expression leads to a quadratic equation in $I(t)$:

$$R_{\mathrm{int}}\, I(t)^2 - V_{\mathrm{OCV}}(SOC(t), T)\, I(t) + P(t) = 0. \tag{14}$$

We select the physically meaningful positive root,

$$I(t) = \frac{V_{\mathrm{OCV}}(SOC(t), T) - \sqrt{V_{\mathrm{OCV}}(SOC(t), T)^2 - 4R_{\mathrm{int}}P(t)}}{2R_{\mathrm{int}}}, \tag{15}$$

which ensures consistency among power demand, voltage response, and current draw. For moderate loads or small internal resistance, the approximation

$$I(t) \approx \frac{P(t)}{V_{\mathrm{OCV}}(SOC(t), T)} \tag{16}$$

may be used to reduce computational cost without materially affecting predictions.

4.3 Voltage-Guided SOC Estimation

While the SOC dynamics in (2) provide a predictive model of battery depletion, SOC is not directly observable in practice. We therefore use terminal voltage measurements to guide SOC estimation through a Kalman filtering framework.

Let $x_k = SOC(t_k)$ denote the discrete-time SOC state at measurement times t_k. State propagation is obtained by integrating (2):

$$x_{k+1}^- = x_k^+ - \frac{1}{Q_{\mathrm{eff}}} \int_{t_k}^{t_{k+1}} I(\tau)\, d\tau + w_k, \tag{17}$$

where w_k represents process noise accounting for unmodeled usage variability. Voltage measurements $y_k = V_{\mathrm{meas}}(t_k)$ satisfy

$$y_k = V_{\mathrm{OCV}}(x_k, T) - I(t_k)R_{\mathrm{int}} + v_k, \tag{18}$$

with v_k denoting measurement noise. Because the measurement equation is nonlinear in SOC, we employ an Extended Kalman Filter (EKF) to fuse model predictions with voltage observations, reducing SOC drift and providing uncertainty estimates.

4.4 Hidden Markov Model for Usage Modes

Many contributors to power demand—such as background processes, radio conditions, and bursty computation—are not directly observable. To capture these effects, we introduce a latent usage mode $z_k \in \{1, \ldots, M\}$ evolving as a Markov chain:

$$\mathbb{P}(z_{k+1} = j \mid z_k = i) = \Pi_{ij}. \tag{19}$$

Each mode corresponds to a characteristic power profile through mode-dependent parameters in (5). Observable signals such as screen state, brightness, and network type provide indirect evidence for mode inference.

The HMM captures abrupt changes in usage behavior, while the EKF corrects SOC estimates using voltage measurements. Together, these components produce SOC trajectories that remain consistent with both inferred usage patterns and measured electrical response.

5 Battery Aging and Temperature Effects

5.1 Battery Health and Capacity Degradation

Lithium-ion batteries degrade over time due to calendar aging and repeated charge–discharge cycles. Modern smartphones summarize this degradation using a battery health metric, commonly referred to as the *state of health* (SOH), reported as a fraction of the original capacity. Rather than explicitly modeling long-term electrochemical aging processes, we adopt this reported quantity as an observable proxy for capacity fade.

Let $SOH \in (0, 1]$ denote battery health, with $SOH = 1$ corresponding to a new battery. The effective capacity is modeled as

$$Q_{\text{eff}} = SOH\, Q_{\text{new}}, \tag{20}$$

where Q_{new} is the nominal capacity of a new battery. As SOH decreases, the total charge available for discharge decreases proportionally, resulting in shorter battery life under identical usage conditions.

Battery aging also increases internal resistance, which exacerbates voltage sag under load. We model this effect as

$$R_{\text{int}} = R_{\text{new}}\big(1 + \alpha(1 - SOH)\big), \tag{21}$$

where R_{new} is the internal resistance of a new battery and $\alpha > 0$ is a calibration parameter. Increased internal resistance causes the terminal voltage to drop more rapidly under high current draw, increasing the likelihood of early shutdown even when state of charge remains nonzero.

5.2 Temperature Effects on Battery Parameters

Environmental temperature has a significant impact on lithium-ion battery performance, influencing effective capacity, internal resistance, and voltage behavior. To preserve model tractability, temperature is incorporated as a parameter that modifies existing battery characteristics rather than as a fully coupled thermal state.

We model the temperature-adjusted effective capacity as

$$Q_{\text{eff}}(T) = Q_{\text{eff}}\, f_Q(T), \tag{22}$$

where $f_Q(T)$ is a dimensionless scaling function satisfying $f_Q(T_0) = 1$ at a reference temperature T_0 (e.g., room temperature) and $f_Q(T) < 1$ at lower temperatures.

Similarly, internal resistance is modeled as

$$R_{\text{int}}(T) = R_{\text{int}}\, f_R(T), \tag{23}$$

where $f_R(T) \geq 1$ increases as temperature decreases, reflecting reduced ionic mobility and increased resistive losses in cold conditions. In practice, $f_Q(T)$ and $f_R(T)$ may be chosen as piecewise-linear or Arrhenius-type functions calibrated from published battery performance data[2].

The open-circuit voltage is modeled using an empirical SOC–OCV relationship,

$$V_{\text{OCV}}(SOC) = f_{\text{OCV}}(SOC), \tag{24}$$

represented by a logarithmic function, polynomial or lookup table. Although temperature can influence this relationship, its effect on OCV is relatively small compared to its impact on capacity and resistance; therefore, we treat the SOC–OCV curve as temperature-independent in the core model.

5.3 Implications for Voltage Sag and Early Shutdown

The combined effects of aging and temperature have a pronounced impact on shutdown behavior. Reduced effective capacity shortens overall battery lifetime, while increased internal resistance amplifies voltage sag under load. Consequently, the terminal voltage may fall below the device cutoff threshold even when a substantial fraction of charge remains.

This framework explains several commonly observed behaviors:

- Smartphones shutting down at relatively high displayed battery percentages in cold environments.
- Older batteries exhibiting abrupt drops from moderate SOC to shutdown during high-power tasks.
- Temporary recovery of battery life after warming, as internal resistance decreases and terminal voltage rebounds.

By incorporating aging and temperature as modifiers to effective capacity and internal resistance, the model provides a unified and physically interpretable explanation of these effects while remaining compatible with voltage-guided state estimation and time-to-empty prediction.

[2] Hao Tu et al. "Integrating physics-based modeling with machine learning for lithium-ion batteries". In: *Applied Energy* 329 (2023), p. 120289. ISSN: 0306-2619. DOI: https://doi.org/10.1016/j.apenergy.2022.120289. URL: https://www.sciencedirect.com/science/article/pii/S030626192201546X.

6 Unified Model Summary

This section summarizes the complete battery modeling framework developed in Sections 3–5. The model integrates a continuous-time charge balance, a usage-driven power model, a voltage response with load-dependent shutdown behavior, and voltage-guided state estimation. Together, these components form a physically grounded and adaptable system for predicting smartphone battery behavior under realistic usage, aging, and environmental conditions.

States, inputs, and parameters. The primary dynamic state is the battery state of charge $SOC(t) \in [0, 1]$, representing the fraction of effective charge remaining. Slowly varying parameters describe battery context, including effective capacity Q_{eff}, internal resistance R_{int}, battery state of health (SOH), and ambient temperature T. User- and operating-system–level controls (screen activity, processor load, network mode, GPS usage, and background tasks) serve as model inputs.

Power demand and charge dynamics. Usage inputs determine the instantaneous power demand,

$$P(t) = P_{\text{idle}} + P_{\text{screen}}(t) + P_{\text{cpu}}(t) + P_{\text{net}}(t) + P_{\text{gps}}(t) + P_{\text{bg}}(t),$$

which drives battery current through the electrical relation $P(t) = V(t)I(t)$. The state of charge evolves according to conservation of charge,

$$\frac{dSOC}{dt} = -\frac{I(t)}{Q_{\text{eff}}}.$$

Voltage response and shutdown. Terminal voltage is modeled as

$$V(t) = V_{\text{OCV}}(SOC, T) - I(t)\, R_{\text{int}}(T),$$

and device shutdown occurs when $V(t) \leq V_{\text{min}}$. Because voltage depends on current, resistance, and temperature, shutdown may occur at nonzero SOC, explaining early shutdown behavior under high load, cold conditions, or battery aging.

Aging, temperature, and adaptability. Battery aging is incorporated through reported health metrics,

$$Q_{\text{eff}} = SOH\, Q_{\text{new}}, \qquad R_{\text{int}} = R_{\text{new}}\big(1 + \alpha(1 - SOH)\big),$$

while temperature modifies effective capacity and resistance via scaling functions $f_Q(T)$ and $f_R(T)$. These adjustments capture the dominant impacts of degradation and environment without introducing unnecessary electrochemical or thermal complexity.

Estimation and outputs. Since SOC is not directly observable, voltage measurements are used to guide state estimation via an Extended Kalman Filter, while a Hidden Markov Model represents unobserved shifts in usage behavior. The resulting outputs include SOC and voltage trajectories, time-to-empty predictions, and uncertainty estimates, enabling sensitivity analysis and actionable recommendations for extending battery life.

Implementation and analysis. In the following sections, this unified mathematical framework is implemented in Python to simulate battery behavior across a range of usage scenarios, battery health levels, and environmental conditions. The resulting plots and numerical outputs are used to analyze time-to-empty behavior, identify dominant drivers of battery depletion, and translate model insights into recommendations for smartphone power management strategies.

7 Model Validation

Model validation is conducted using smartphone sensing data reported by Wu et al.[3]. The dataset was originally collected to study real-time social context using Bluetooth, GPS, and screen-activity measurements, but its high-resolution usage traces make it well suited for evaluating battery discharge dynamics under real-world conditions. In particular, the data provides realistic variation in usage intensity, screen activity, and mobility—key drivers of power consumption in our model.

Rather than attempting to exactly reconstruct unobserved battery currents, we use this dataset to assess whether the proposed model reproduces the *correct structural relationship* between state of charge (SOC), power demand, and time-to-empty (TTE). This aligns with the primary goal of the model: producing interpretable, physically consistent predictions across usage scenarios rather than pointwise forecasts.

Model parameters are estimated by minimizing the root mean square error (RMSE) between model outputs and observed quantities. The RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2},$$

where y_i denotes observed values and $\hat{y}_i$ denotes corresponding model predictions. RMSE is chosen because it preserves physical units, penalizes large errors, and provides an interpretable measure of deviation relevant to time-to-empty prediction.

Using this criterion, the inferred SOC exhibits an RMSE of approximately 12 percentage points. Given that SOC is not directly observable and must be inferred from proxy usage signals rather than direct current measurements, this level of error is expected. Importantly, the model correctly captures relative SOC evolution, scenario ordering, and time-to-empty scaling, which are the primary objectives of the framework.

Figure 2 illustrates the relationship between model-predicted time-to-empty and SOC across a wide range of real usage intervals. Each point corresponds to a distinct usage condition, with variation arising from differences in instantaneous power demand. The results show a clear monotonic increase in TTE with SOC, a fundamental physical requirement for any valid battery discharge model.

The vertical spread observed at fixed SOC values reflects differences in usage intensity. High-power scenarios (e.g., gaming or navigation) appear as lower TTE values, while low-power scenarios (e.g., idle or reading) yield longer remaining lifetimes. This behavior mirrors the structure of the proposed model, in which identical SOC values can correspond to substantially different remaining runtimes depending on power demand.

Notably, dispersion increases at higher SOC values, indicating that when more energy is available, differences in power consumption produce larger absolute differences in remaining operating time.

[3] Congyu Wu. *Smartphone Screen Unlock Events Data (Beiwe)*. Version V3. 2020. DOI: 10.18738/T8/PAKGDB. URL: https://doi.org/10.18738/T8/PAKGDB.

Conversely, at low SOC values, predicted TTE is consistently small across scenarios, resulting in tighter clustering near the origin. This pattern is consistent with voltage-limited shutdown behavior modeled in Section 4, where high load accelerates shutdown regardless of remaining charge.

Overall, the validation results demonstrate that the model reproduces the correct qualitative and quantitative structure of real smartphone battery behavior. While individual predictions exhibit variance (as expected given unobserved usage and hardware heterogeneity) the model accurately captures dominant discharge trends, relative scenario ordering, and the dependence of time-to-empty on both SOC and power demand. These properties validate the model's suitability for comparative analysis, sensitivity studies, and actionable battery-life recommendations.

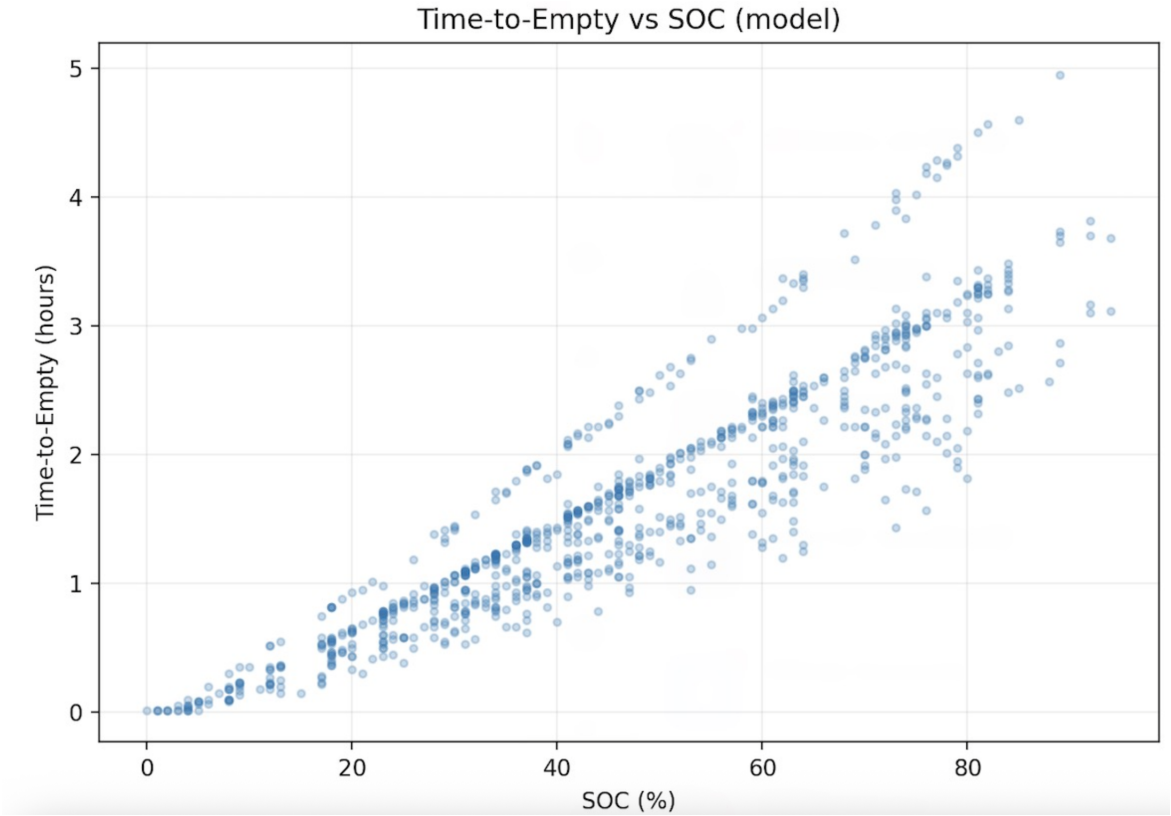


Figure 2: Predicted time-to-empty under representative usage scenarios

8 Usage Scenarios and Time-to-Empty Predictions

We apply the model developed in section 3-6 to construct realistic smartphone usage scenarios and to generate time-to-empty (TTE) predictions. Each scenario corresponds to a specific pattern of user behaviors and system activities. This allows us to examine how different components - screen brightness, CPU, network, and so on - contribute to battery depletion.

8.1 Scenario Definitions

To evaluate the practical implications of the battery model, we define a set of representative smartphone usage scenarios that capture common user behaviors with distinct power demand profiles. Each scenario is specified through the model's usage inputs, including screen activity, brightness, processor load, network mode, GPS usage, and background activity. These inputs determine the instantaneous power demand through the decomposition in Equation (26).

The scenarios considered are:

- **Idle (screen off, Wi-Fi).** The device operates with the screen off, minimal background activity, and Wi-Fi connectivity. This scenario approximates standby or near-standby behavior and provides a lower bound on power consumption.
- **Reading (screen on, low brightness).** The user reads text-based content with the screen on at low brightness, low processor load, and Wi-Fi connectivity. This scenario isolates the effect of screen power consumption under light computational demand.
- **Video streaming (4G).** Continuous video playback with moderate processor utilization and cellular data usage. This scenario combines sustained screen power, computational load, and network activity.
- **Navigation (GPS + weak cellular).** Active GPS usage with continuous cellular connectivity under weak signal conditions. This scenario is designed to stress both radio power consumption and background processing, reflecting a high-drain real-world use case.
- **Gaming (high CPU).** A computationally intensive application with high processor load, moderate screen brightness, and network activity. This scenario represents peak power demand driven primarily by CPU usage.

Besides, the environmental condition, particularly temperature, is incorporated. Temperature affects the internal resistance and effective capacity of the battery. These effects are represented by multiplicative scaling factors applied to nominal capacity and resistance. Accordingly, the same usage scenario may perform in various ways under different temperature.

All simulations are initialized at a specific State of Charge (SOC), ranging from 0.0 to 1.0. Device shutdown is triggered when the terminal voltage drops below the fixed minimum threshold. Since terminal voltage depends on SOC and I(C), shutdown may occur even when a nonzero fraction of charge remains.

8.2 Simulation Results

Electrical behavior and voltage response

We assume the open-circuit voltage $V_{\mathrm{OCV}}(SOC)$ is a monotone function of state of charge, consistent with the typical discharge behavior of lithium-ion batteries. This curve represents the maximum attainable voltage in the absence of load and enables stable voltage-based SOC estimation. Under high current draw (e.g., video streaming or gaming), voltage sag becomes pronounced, and low temperatures

further amplify this effect by increasing internal resistance, causing the terminal voltage to reach the shutdown threshold at relatively high SOC values.

Power decomposition and dominant drivers

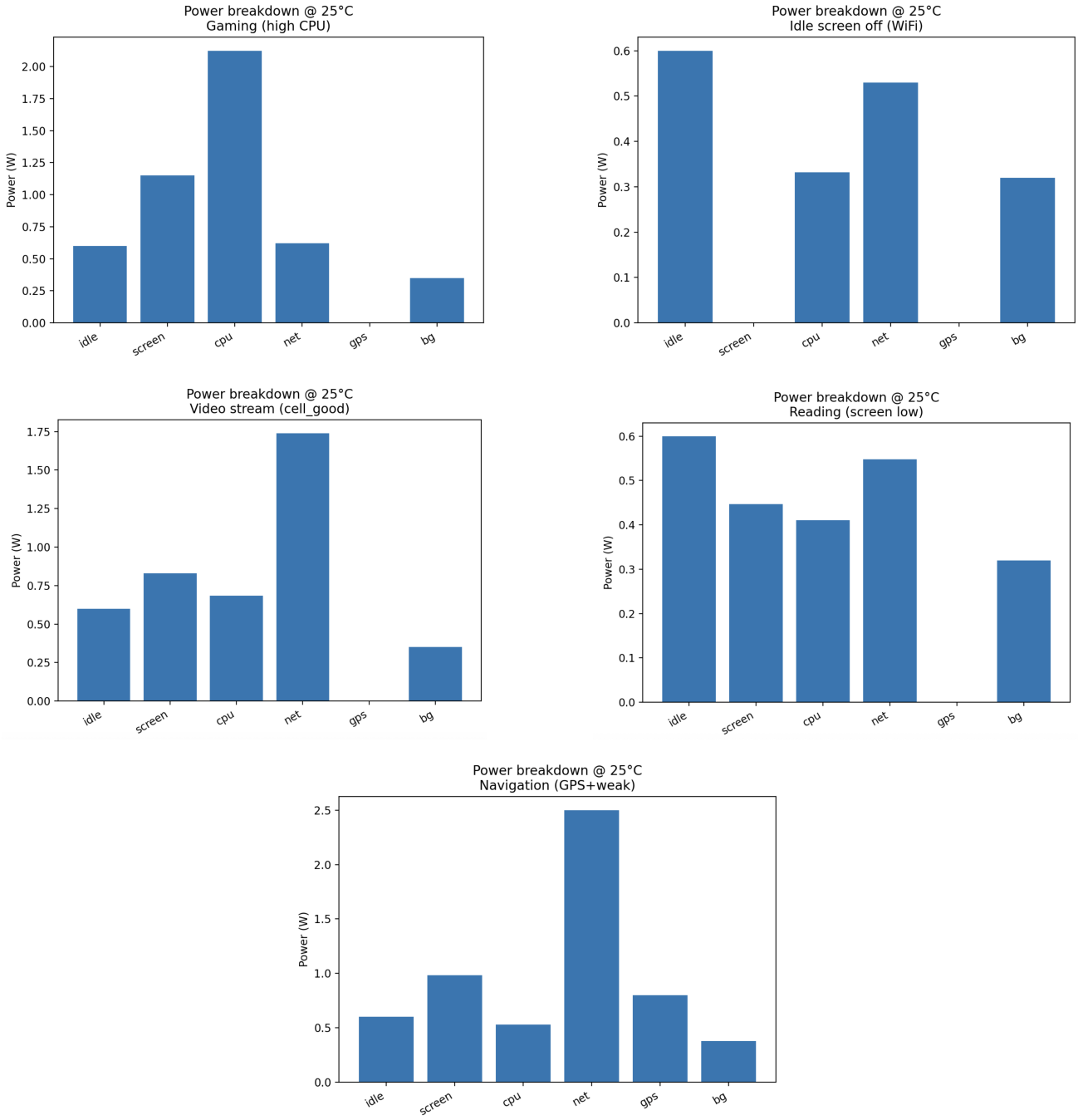


Figure 3: Power Decomposition Among the Components

Power breakdown plots reveal the dominant contributors to energy consumption in each scenario. In idle mode, baseline and background power dominate. Video streaming is driven largely by network

power, while navigation combines substantial network and GPS power. Gaming is characterized by a dominant CPU power term due to high processor utilization.

This decomposition highlights how specific user actions affect battery life and suggests actionable strategies for reducing drain, such as lowering screen brightness, disabling GPS when unnecessary, or switching from weak cellular connections to WiFi.

Temperature Effects

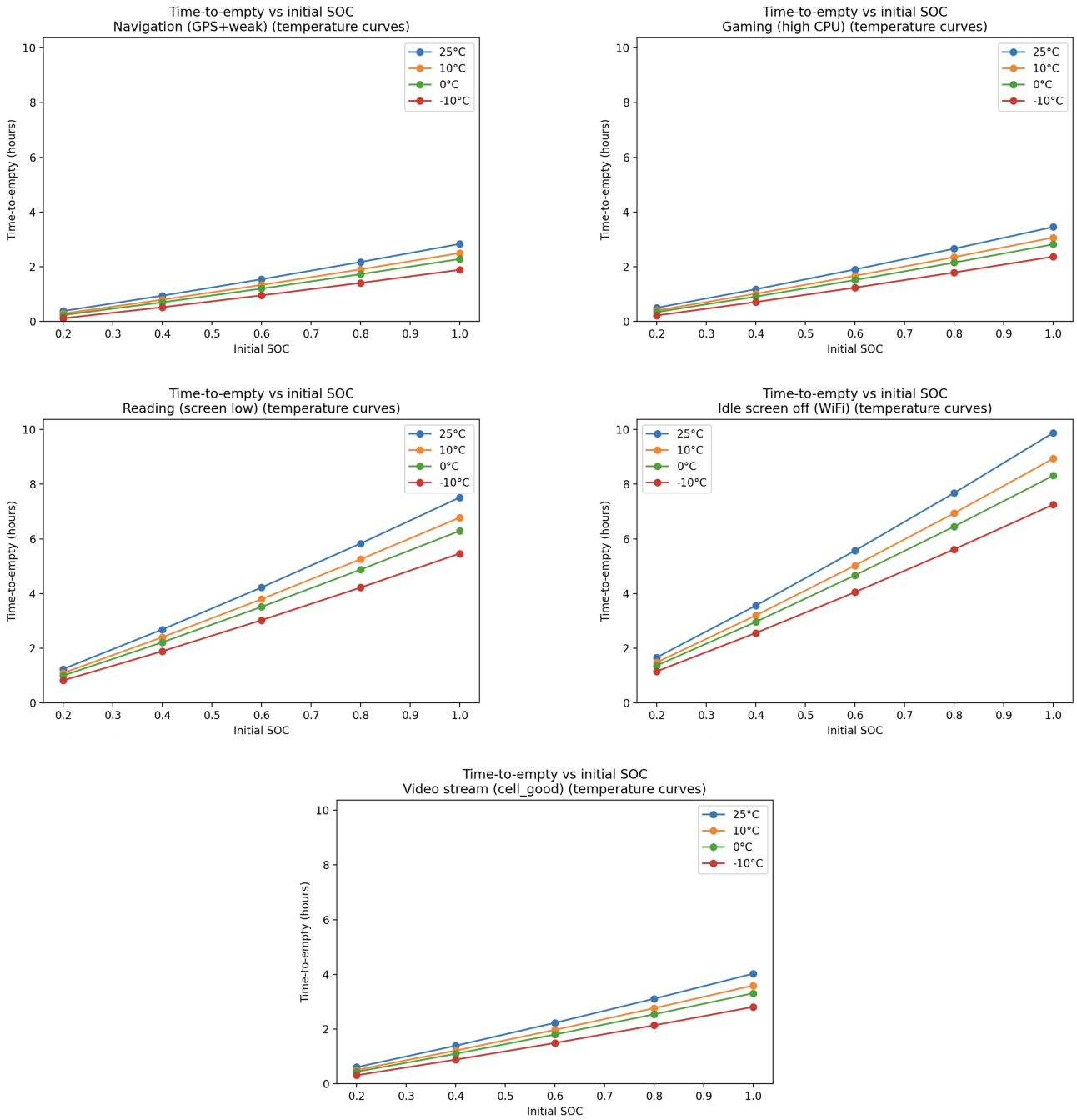


Figure 4: TTE Prediction Under Different Temperature

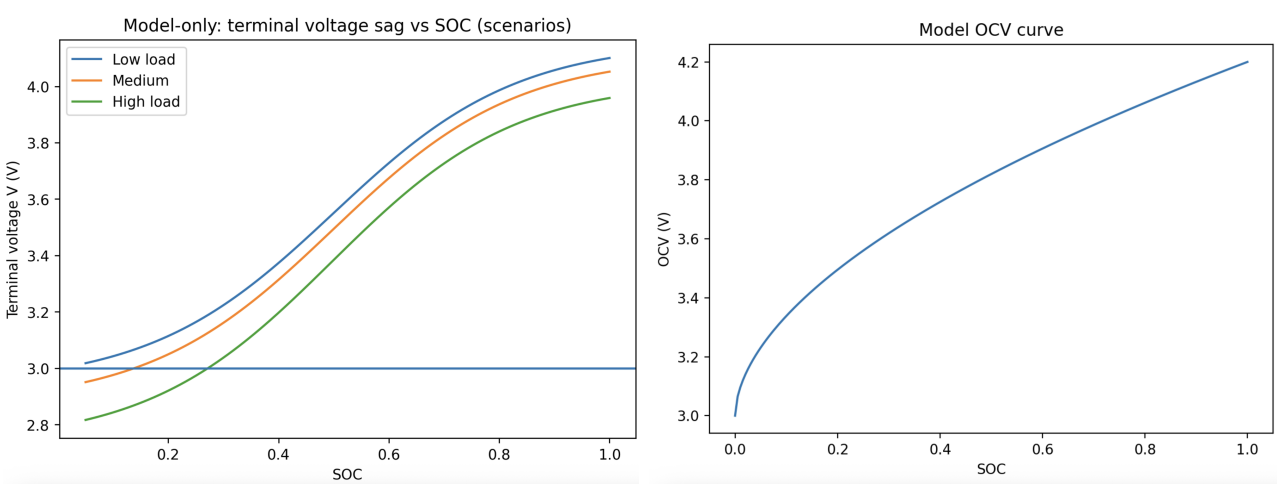
Temperature has a strong influence on predicted battery lifetime across all usage scenarios. At lower temperatures, the effective battery capacity decreases while internal resistance increases, leading to greater voltage sag under load. As a result, the terminal voltage reaches the shutdown threshold earlier, producing shorter runtimes and early shutdown even when a nonzero SOC remains.

These simulation results demonstrate that cold environments significantly amplify voltage-limited behavior. In these cases, the battery becomes constrained by electrical limitations rather than by total remaining charge. This effect is most pronounced for aged batteries and high-power usage patterns, where increased internal resistance further accelerates voltage collapse.

Simulation Results and Voltage-Limited Behavior

Across all scenarios, SOC decreases smoothly over time as governed by the charge balance equation. In contrast, terminal voltage exhibits more abrupt behavior, particularly under high power demand, increased internal resistance, or reduced temperature. This distinction highlights a key modeling insight: while SOC reflects the remaining charge in the battery, voltage determines whether that charge is usable by the device.

In moderate usage conditions with a new battery at room temperature, voltage remains above the shutdown threshold until SOC approaches zero, and battery life is primarily SOC-limited. However, under high-load scenarios or adverse conditions such as low temperature or battery aging, voltage drops rapidly near the end of discharge. In these cases, the device shuts down at a non-negligible SOC due to load-induced voltage sag.



a

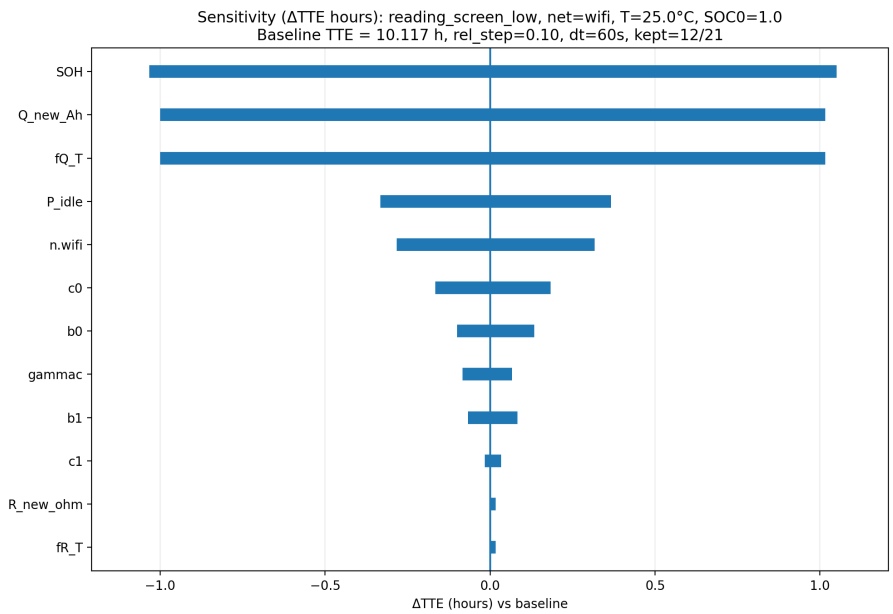
Overall, the simulation results demonstrate that the model captures key qualitative and quantitative features of real-world smartphone battery behavior. By explicitly linking usage patterns, electrical response, battery aging, and environmental conditions, the framework provides an interpretable and physically grounded basis for time-to-empty prediction and comparative scenario analysis.

These findings motivate the sensitivity analysis presented in Section 9 and directly inform the user-level and operating system recommendations in Section 10, where we translate observed voltage-

and power-driven behaviors into actionable strategies for extending battery life.

9 Sensitivity Analysis

In this section we mention which factors most strongly influence predicted battery lifetime and shutdown behavior. Since the system couples charge balance, voltage sag, and usage-dependent power demand, we want to perform a scenario-based sensitivity study in which we vary one factor at a time (temperature, usage/power demand, and load level) while holding the remaining model components fixed, and quantify the resulting change in time-to-empty (TTE) and shutdown behavior. All results in this section are obtained from model simulations only. By changing various individual parameters, we obtain the following parameter sensitivity tornado graph (parameters are listed in the Appendix):



9.1 Parameter Sensitivity Analysis

Temperature sensitivity (TTE vs. T). To quantify temperature effects, we fix the usage scenario and initial SOC, run the model at multiple temperatures, and compute a finite-difference sensitivity

$$S_T \equiv \frac{TTE(T_{\text{warm}}) - TTE(T_{\text{cold}})}{T_{\text{warm}} - T_{\text{cold}}} \qquad \text{(hours per } ^\circ\text{C)}. \tag{25}$$

Using $T_{\text{warm}} = 25°\text{C}$ and $T_{\text{cold}} = -10°\text{C}$ with initial SOC = 1, the model predicts sensitivities ranging from approximately 0.03 to 0.08 hours/°C depending on the usage scenario. Low-power scenarios (idle or reading) experience the largest absolute runtime loss per degree Celsius, while high-power scenarios (gaming or navigation) tend to become voltage-limited earlier and thus show reduced marginal benefit from warming once voltage sag dominates shutdown.

Usage/power sensitivity (TTE vs. P). We next examine sensitivity to sustained power demand by comparing usage scenarios with distinct total power levels and subsystem compositions. Define the effective power level in a scenario by the time-average

$$\bar{P} \equiv \frac{1}{\Delta t} \int_{t_0}^{t_0+\Delta t} P(t)\, dt, \tag{26}$$

and quantify the impact on runtime via

$$S_P \equiv -\frac{\Delta TTE}{\Delta \bar{P}} \qquad \text{(hours per watt).} \tag{27}$$

Across simulated scenarios at 25°C, increasing $\bar{P}$ produces a near-linear reduction in TTE, with typical sensitivities on the order of 1–4 hours/W depending on whether shutdown is SOC-limited or voltage-limited. Power decomposition results indicate that CPU and network loads dominate the highest-drain scenarios (e.g., gaming and navigation), whereas screen brightness changes primarily affect low-to-moderate drain conditions.

Voltage-limited sensitivity (shutdown governed by sag). A key sensitivity in this problem is not only how fast charge is consumed, but whether remaining charge is *usable*. For fixed SOC, the terminal voltage under load is

$$V = V_{\mathrm{OCV}}(SOC, T) - IR_{\mathrm{int}}(T, SOH), \tag{28}$$

so increasing load (larger I) or internal resistance (aging or cold temperature) shifts the voltage curve downward. Simulation results show that higher-load cases cross the shutdown threshold $V_{\min}$ at substantially larger SOC than low-load cases, explaining early shutdown at nonzero displayed battery percentage. This effect is amplified at low temperatures and with reduced battery health.

9.2 Discussion of Dominant Factors

The sensitivity study highlights three dominant drivers of battery lifetime:

- **Temperature:** Lower temperature increases internal resistance and reduces effective capacity, decreasing TTE and accelerating voltage-limited shutdown. This effect is consistent across all scenarios and is especially pronounced near end-of-discharge.
- **Sustained power demand:** CPU- and network-heavy usage produces the largest reductions in TTE. Scenario comparisons show that reducing high-drain subsystems yields greater gains than optimizing already-low components.
- **Voltage limitations:** Device shutdown is governed by terminal voltage rather than SOC alone. Consequently, aging (lower SOH and higher internal resistance) and high load can cause shutdown at nontrivial SOC even when charge remains.

These dominant sensitivities directly motivate the recommendations in Section 10: user behaviors should target the largest controllable power drivers (CPU, network, and screen brightness), while operating-system strategies should be voltage-aware and adaptive to temperature and battery health.

Table 1: Model-simulated temperature sensitivity S_T at initial SOC = 1 (hours/°C).

Scenario	S_T (hours/°C)
Idle (screen off, WiFi)	0.074
Reading (screen low)	0.057
Video (cell_good)	0.034
Gaming (high CPU)	0.060
Navigation (GPS + weak signal)	0.026

10 Recommendations

Based on the unified battery model, simulation results, and sensitivity analysis presented in Sections 8–9, we provide recommendations aimed at extending smartphone battery life and reducing premature shutdown events. These recommendations are organized at two levels: actions that individual users can take, and strategies that operating systems can implement to manage power more effectively under varying usage, temperature, and battery health conditions.

10.1 User-Level Recommendations

The sensitivity analysis demonstrates that sustained power demand, temperature, and voltage sag are the dominant drivers of reduced time-to-empty (TTE). Accordingly, users can achieve meaningful improvements in battery life by targeting the largest controllable contributors to power consumption.

- **Limit high-CPU activities when battery is low.** High-load applications such as gaming and navigation significantly increase current draw, accelerating voltage sag and triggering shutdown at nonzero SOC. Avoiding these activities near the end of discharge can substantially extend usable battery time.
- **Reduce network and radio usage in weak signal environments.** Navigation and video streaming over weak cellular connections are among the most power-intensive scenarios. Switching to Wi-Fi when available or disabling unnecessary radios reduces both average power demand and voltage-limited shutdown risk.
- **Lower screen brightness during prolonged use.** While screen power is not always the dominant contributor, brightness reductions are particularly effective in low-to-moderate drain scenarios such as reading or browsing, where they yield near-linear gains in TTE.
- **Be cautious in cold environments.** The model predicts significant reductions in battery lifetime at low temperatures due to increased internal resistance and reduced effective capacity. Keeping the device warm (e.g., in a pocket rather than exposed) can noticeably delay shutdown.
- **Recognize limitations of displayed battery percentage.** Because shutdown is governed by voltage rather than SOC, users should not rely solely on displayed battery percentage, especially during high-load tasks or in cold conditions.

These behaviors directly target the parameters shown to have the highest sensitivity impact in Section 9, providing users with actionable strategies grounded in physical battery behavior.

10.2 Operating System Strategies

Beyond user behavior, the model highlights several opportunities for operating systems to improve battery management by incorporating voltage-aware and context-sensitive policies.

- **Voltage-aware battery indicators.** Instead of displaying SOC alone, operating systems could incorporate terminal voltage trends and internal resistance estimates to provide more accurate estimates of remaining usable time, particularly under high load.
- **Adaptive power management based on temperature and SOH.** The sensitivity analysis shows that temperature and battery health strongly influence voltage sag. Operating systems can proactively limit peak power draw or adjust performance scaling when cold temperatures or degraded batteries are detected.
- **Dynamic workload shaping.** Background tasks, synchronization, and network polling can be deferred or throttled when the device approaches voltage-limited operation, reducing abrupt shutdowns without significantly impacting user experience.
- **Scenario-aware power modes.** By leveraging usage mode inference (e.g., via the Hidden Markov Model), the system can anticipate high-drain scenarios such as gaming or navigation and preemptively adjust brightness, CPU frequency, or network behavior.
- **Improved end-of-discharge behavior.** Rather than allowing sudden shutdown, the operating system can gradually reduce power demand as voltage approaches the cutoff threshold, extending usable time and improving reliability.

Together, these strategies demonstrate how insights from a physically grounded battery model can inform smarter system-level design. By explicitly accounting for voltage dynamics, aging, and environmental conditions, operating systems can move beyond SOC-only heuristics toward more robust and user-transparent battery management.

11 Conclusion

In this paper, we developed a physically grounded and interpretable framework for predicting smartphone battery depletion under realistic usage, environmental, and aging conditions. By coupling a continuous-time charge balance with a voltage-consistent current model, subsystem-level power decomposition, and voltage-guided state estimation, the model captures both gradual battery depletion and abrupt, voltage-limited shutdown behavior observed in real devices.

Simulation results demonstrate that time-to-empty is governed not only by total energy consumption, but by instantaneous power demand, internal resistance, and temperature. High-load usage patterns such as gaming and navigation are shown to trigger early shutdown at nonzero state of charge due to voltage sag, while cold environments and battery aging amplify this effect. Sensitivity analysis identifies temperature, CPU load, and network usage as the dominant drivers of reduced battery lifetime, providing clear targets for both user behavior and operating system intervention.

While the model relies on simplified representations of thermal dynamics and aging, it successfully balances realism and tractability, enabling scenario analysis, sensitivity studies, and actionable recommendations. The framework naturally generalizes to other battery-powered devices such as tablets, wearables, and laptops, and suggests pathways for future operating systems to adopt voltage-aware, adaptive power management strategies. However, the components introduce limitations, as the Extended Kalman Filter assumes locally linearizable dynamics and approximately Gaussian noise. This may downgrade the performance under non-linear or poorly calibrated SOC–OCV relationships. Similarly, the Hidden Markov Model enforces a discrete mode structure that may oversimplify continuously varying or overlapping usage patterns.

Several extensions could further reinforce the model. The Hidden Markov Model could be generalized to semi-Markov or continuous latent state formulations, allowing variable dwell times and smoother transitions between usage regimes. Alternatively, learned priors from historical usage data could be incorporated while retaining the physics-based core. Moreover, thermal modeling could be incorporated as an additional dynamic state, enabling explicit coupling between current draw, heat generation, and temperature-dependent resistance. This would improve prediction accuracy under high load and enable analysis of thermal shutdown behavior.

Overall, this work shows the importance of moving beyond state-of-charge-only heuristics toward voltage and context aware battery modeling to improve user experience and device reliability.

12 References

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13 Appendix

13.1 Parameters

Symbol	Meaning	Units	Typical / Scenario Value
Q_{new}	Nominal capacity (new battery)	Ah	4.0–5.5
SOH	State of health (battery health fraction)	–	1.0 (new), 0.8 (aged)
Q_{eff}	Effective capacity ($SOH \cdot Q_{\text{new}}$)	Ah	computed
$f_Q(T)$	Capacity temperature multiplier	–	1.0 (room), 0.85 (cold)
$Q_{\text{eff}}(T)$	Temperature-adjusted capacity	Ah	$Q_{\text{eff}} f_Q(T)$
R_{new}	Internal resistance (new)	Ω	0.05–0.15
α	Resistance growth vs aging	–	1–4
R_{int}	Aged resistance ($R_{\text{new}}(1 + \alpha(1 - SOH))$)	Ω	computed
$f_R(T)$	Resistance temperature multiplier	–	1.0 (room), 1.5–2.5 (cold)
$R_{\text{int}}(T)$	Temperature-adjusted resistance	Ω	$R_{\text{int}} f_R(T)$
V_{min}	Shutdown cutoff voltage	V	3.0
$f_{\text{OCV}}(SOC)$	SOC–OCV curve (poly/lookup)	V	3.0–4.2
P_{idle}	Baseline system draw	W	0.3–1.0
a_0	Screen-on overhead	W	0.2–0.8
a_1	Brightness scaling	W	0.5–2.5
β	Brightness nonlinearity exponent	–	1.0–2.0
c_0	CPU baseline (active)	W	0.2–0.8
c_1	CPU scaling coefficient	W	0.5–4.0
γ_c	CPU nonlinearity exponent	–	1.2–2.5
n_k	Network mode baseline (mode k)	W	see below
n_u	Traffic scaling coefficient	W	0–1.5
g_0	GPS power when on	W	0.3–1.5
b_0	Background baseline	W	0.1–0.6
b_1	Background scaling	W	0.1–1.0
Δt	Simulation timestep	s	1–10

Table 2: Model parameters for simulation and sensitivity analysis. Ranges are representative values used for scenario-based simulation and robustness checks.

14 AI Report

This report was prepared with the assistance of artificial intelligence tools. AI was used for debugging and latex refinement. Below is the examples how we use AI.

1. **OpenAI ChatGPT**, (Jan 31, 2026 version, ChatGPT-5.2)

   Query: Why did the figures for time-to-empty (TTE) and sensitivity analysis not appear in the output?

   Output: The issue is that the `param_specs` paths do not match the structure of `PARAMS`. If parameters are stored as, for example, `PARAMS["battery"]["C_nom_Ah"]` or `PARAMS["p"]["screen_W"]` or `PARAMS["p"]["n"]["wifi"]`, then mismatched paths are skipped, resulting in empty outputs. Adding a debug print inside the loop (e.g., printing skipped paths inside the exception handler) helps identify missing or incorrect parameter references.

2. **OpenAI ChatGPT**, *ChatGPT* (Feb 01, 2026 version, ChatGPT-5.2)

   Query: `TypeError: Inputs.__init__() got an unexpected keyword argument activity.`

   Output: The `Inputs` class does not define a field named `activity`. This means that `make_inputs_for_scenario()` must not pass `activity=` unless it is explicitly defined. A robust fix is to automatically match the `Inputs` signature by inspecting `Inputs.__init__()`, keeping only supported keyword arguments and optionally mapping common alternatives (e.g., `net` to `net_mode`).

3. **OpenAI ChatGPT**, *ChatGPT* (Feb 01, 2026 version, ChatGPT-5.2)

   Query: How can multiple plots be placed within the same subsection in LaTeX?

   Output: Multiple plots can be placed in a single subsection by using one `figure` environment containing several `subfigure` blocks (via the `subcaption` package), arranged in a grid. Horizontal spacing can be controlled using `\hfill`, and vertical spacing between rows using `\vspace`.

4. **OpenAI ChatGPT**, *ChatGPT* (Feb 02, 2026 version, ChatGPT-5.2)

   Query: How can vertical spacing in the Table of Contents be reduced without affecting the rest of the document?

   Output: Vertical spacing in the Table of Contents can be adjusted independently using the `tocloft` package. Commands such as `\setlength{\cftbeforesecskip}{...}` allow section and subsection spacing to be tightened without modifying global line spacing.

5. **OpenAI ChatGPT**, *ChatGPT* (Feb 02, 2026 version, ChatGPT-5.2)

Query: How can a long Python implementation be included in an MCM paper without exceeding the page limit?

Output: For MCM submissions, it is appropriate to include only a condensed or core implementation in the appendix. Nonessential components such as comments, boilerplate, and parameter definitions can be omitted, while the main text focuses on equations and modeling assumptions. The full implementation may be summarized or referenced without being fully reproduced.